Bocconi

# CORPORATE FINANCE 30017

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## Firm's Financial Data

Università Bocconi
MILANO

# Agenda

PREPARING FOR DATA WORK

INTRODUCTION TO WRDS

OBTAINING STOCK RETURNS

CLASS ROOM EXERCISE:
INDUSTRY CHARACTERISTICS

# PREPARING FOR DATA WORK

# Data relevant for Corporate Finance

- Firm Financial statements
  - Balance sheets, income statements, cash flow statements
  - Typically annual frequency or quarterly frequency
  - Publicly listed vs. private firms, US, global,…
- Financial contracts
  - Loans (type, volume, rates, covenants, syndication…)
  - Bonds (seniority, face value, coupon structure, covenants…)
  - Derivatives (options, CDS…)
- Market prices and transactions (exchanges vs. OTC)
- Data characterising bankruptcies, default and restructuring events
- Investor/creditor portfolios (investor-security-time data)
- Structured data on executive biographies and careers
- …

# A word of caution

*"Not everything that can be measured is important*
*—and not everything important can be measured."*

(attributed to A. Einstein)

— Also see: McNamara Fallacy and related psychological research

- People tend to focus excessively on quantitative metrics (more salient)

# Identifiers

- Fundamental to any (financial) data analysis: **What is the unit of observation***?*
- Each data set is indexed by a selected set of variables
  - *E.g.,* ***Firm*** *X Quarter, or* ***Firm*** *X* ***Security*** *X Day, or* ***Bank*** *X* ***Borrower*** *X Month*
  - How can we ***uniquely*** **identify institutions and contracts** across time and datasets?
    - It is complicated, often no harmonized approach
    - "Header" vs. "historical" identifiers? (Firms may under M&A)
    - Consolidated vs. unconsolidated?

# Identifiers you will likely encounter

- Firm Identifiers:
  - Company Name, Ticker (general but to be used with caution, lack of standardization)
  - Compustat GVKEY (for any data coming from Standard & Poors)
  - CIK (to identify company in filings with the SEC, USA)
  - SEDOL (for UK firms)
  - Bureau van Dijk ID (for private firms, focus on Europe)
- Security Identifiers:
  - ISIN (for stocks around the world)
  - CUSIP (for US securities; can also be used to identify issuing firm)
  - PERMNO (CRSP identifiers for US publicly listed stocks)
  - IBES Ticker (for analyst estimates)

# Company names and tickers

- Both are character strings
- Company names:
  - Often abbreviated or manipulated by the data vendors causing different variations
- Tickers:
  - Usually 3 or 4 characters
  - Widely used and frequently recognizable BUT:
    - Can change at any time and can be reused by a different company
    - One company can have multiple tickers, e.g., Bloomberg “ticker”, NYSE “ticker”, I/B/E/S “ticker”
- In short, names and tickers are terrible identifiers…
  - If necessary, use fuzzy string matching algorithms

| ENTITY_NAME | TICKER |
|---|---|
| APPLE INC | AAPL |
| DEERE & CO | DE |
| MACY'S INC | M |
| MICROSOFT CORP | MSFT |

# Compustat GVKEY

- Identifies a company
  - Is mapped to the most recent ticker, company name, etc.
- Usually a 6 digit number (if less = leading zeros)
- Does not change through the life of a company (company death = liquidation or acquisition)
- COMPUSTAT covers (essentially) all US corporations that file with the SEC
  - Filing with SEC required for issuing tradable securities (stocks, bonds)
  - In terms of value (numbers), majority of US firms do (not) show up there
  - Also global module for non-US public firms
- In short, GVKEY is a very good identifier

| ENTITY_NAME | GVKEY |
|---|---|
| VERIZON COMMUNICATIONS INC | 002136 |
| HERSHEY CO | 005597 |
| TEAVANA HOLDINGS INC | 187119 |

# CIK: Central Index Key

- Used to identify corporations (and individuals) who filed disclosures with the SEC
- Ten digit number
- Never reused
- CIK, in short, is a great identifier – for US firms!

- https://www.sec.gov/edgar/search/

| ENTITY_NAME | CIK |
|---|---|
| DISNEY (WALT) CO | 0001001039 |
| TWITTER INC | 0001418091 |
| SUNGLASS HUT INTL INC | 0000899240 |

# CRSP PERMNO (and PERMCO)

- PERMNO identifies securities.
  - 5 digit number
  - Permno is mapped to historical tickers, company names, etc., with start and end dates in an event file.
- PERMCO identifies companies.
- Permno and Permco do not change when a company changes their name or ticker.
- Never reassigned
- In short, PERMNO and PERMCO are perfect identifiers (thx @Booth)

| ENTITY_NAME | PERMNO | PERMCO | FIRST_DATE | LAST_DATE |
|---|---|---|---|---|
| APPLE COMPUTER INC | 14593 | 7 | 1980-12-12 | 2007-01-10 |
| APPLE INC | 14593 | 7 | 2007-01-11 | 2013-12-31 |
| TESLA MOTORS INC | 93436 | 53453 | 2010-06-29 | 2013-12-31 |

# CUSIP (Committee on Uniform Securities Identification Procedures)

- Widely used in databases to identify securities in USA and Canada
- Changes infrequently and never reused
- Not free, CUSIP license to be purchased by licensee company
- Can be 6, 8, or 9 alpha-numeric characters
  - First 6 characters identify the issuer (firm)
  - Seventh and eighth identify the issue (security)
  - Ninth character is a check digit
- CUSIP, in short, very good but imperfect identifier

| ENTITY_NAME | CUSIP_FULL |
|---|---|
| YAHOO INC | 984332106 |
| CITYBANK | 17770A109 |
| FACEBOOK INC | 30303M102 |

# ISIN (International Securities Identification Number)

- Used for identifying securities from all countries
- Consists of 12 alphanumeric characters
  - The first 2 characters identify country of origin.
  - The 3rd to 11th characters are unique characters.
    - For USA, these correspond to the CUSIP identifier.
  - The 12th character is a check character.
- But, what happens when a company changes listing/merges/re-lists/etc..?
  - NL0010877643 is FIAT, for example…
- ISIN, in short, is a so-so identifier

| ENTITY_NAME | ISIN |
|---|---|
| KENYA POWER & LGHT | KE0000000349 |
| BRAZIL HOSPITALITY GROUP SA | BRBHGRACNOR9 |
| THAI TAP WATER SUPPLY PCL | TH0961010004 |

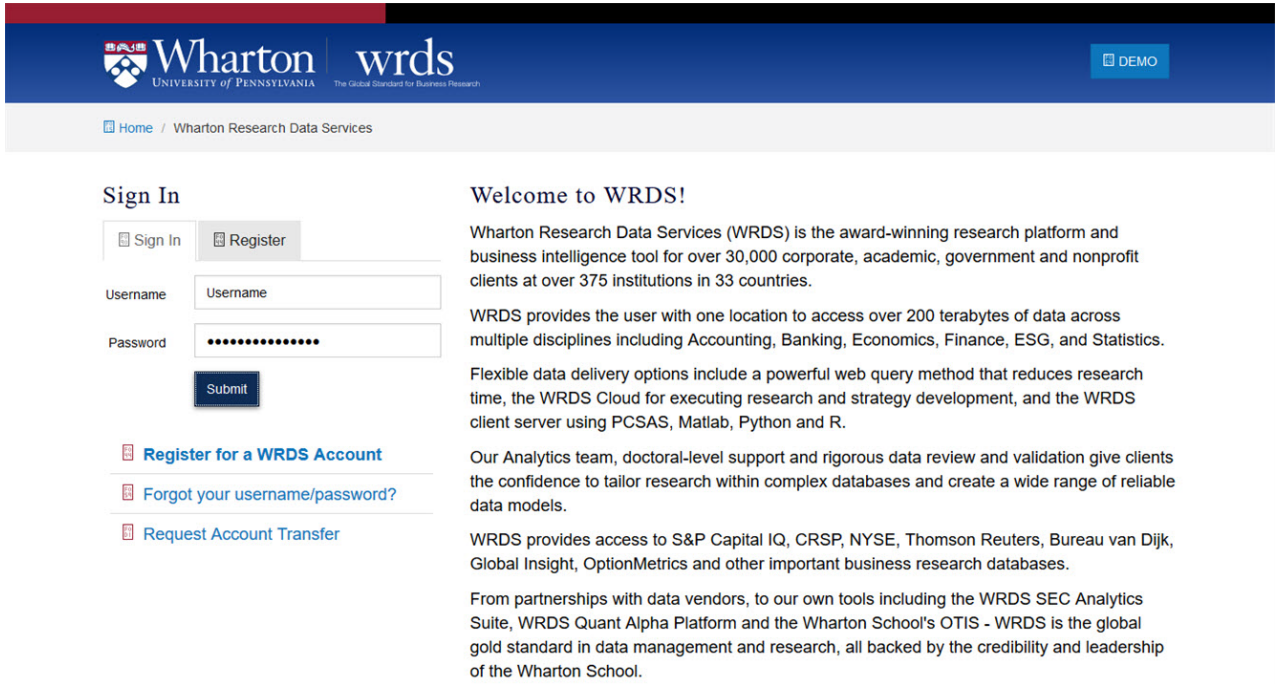
# INTRODUCTION TO WRDS

# Wharton Research Data Services (WRDS)

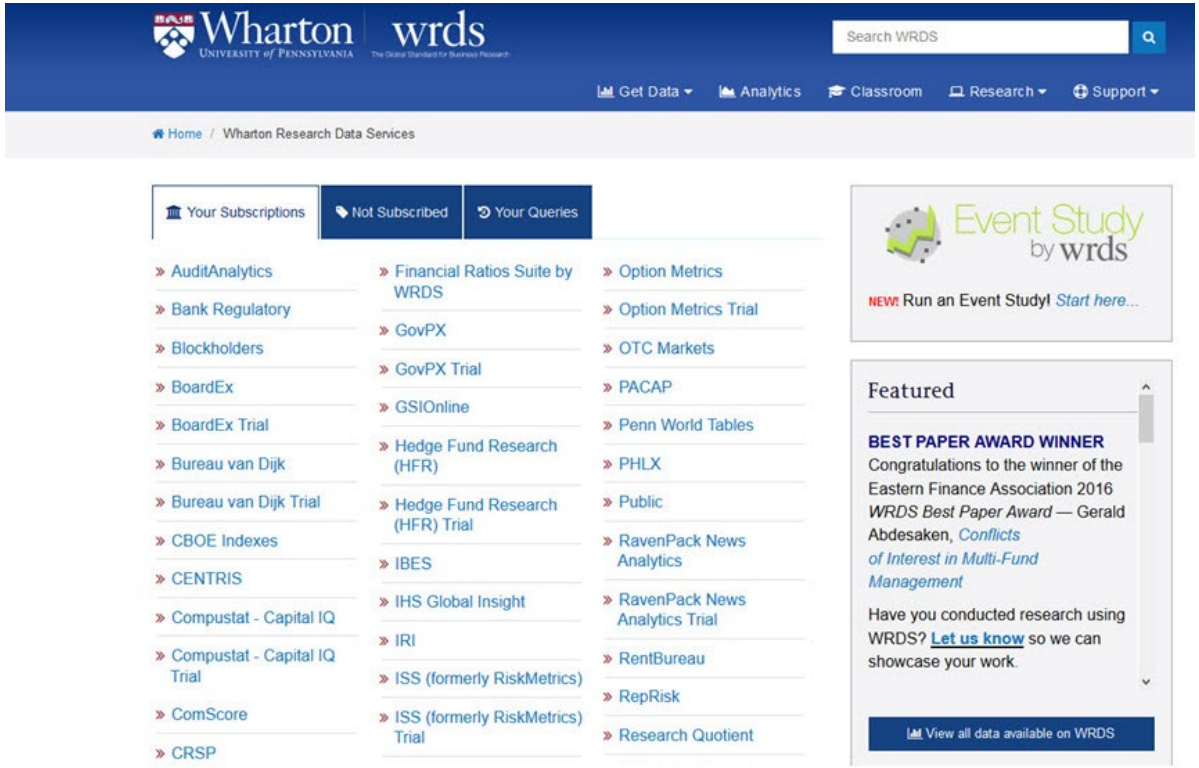
- Hosted at the University of Pennsylvania, Wharton School of Business
  - Since 1993: effort to collect, harmonize, link and distribute research-grade financial data
- WRDS is Internet-based with multiple access options
  - https://wrds-www.wharton.upenn.edu/
  - Web queries: easy-to-use, point-and-click interface [THIS IS WHAT WE WILL USE]
  - PC SAS CONNECT remote libraries [REQUIRES A "THESIS"-TYPE ACCOUNT]
  - UNIX X-Windows and batch job power [REQUIRES A "THESIS"-TYPE ACCOUNT]

- *Assume we wanted to analyze monthly stock prices for US companies over the past 20 yrs…*

# WRDS Login


Wharton
UNIVERSITY of PENNSYLVANIA
wrds
The Global Standard for Business Research
DEMO
Home / Wharton Research Data Services
Sign In
Sign In
Register
Username
Username
Password
Submit
Register for a WRDS Account
Forgot your username/password?
Request Account Transfer
Welcome to WRDS!
Wharton Research Data Services (WRDS) is the award-winning research platform and business intelligence tool for over 30,000 corporate, academic, government and nonprofit clients at over 375 institutions in 33 countries.
WRDS provides the user with one location to access over 200 terabytes of data across multiple disciplines including Accounting, Banking, Economics, Finance, ESG, and Statistics.
Flexible data delivery options include a powerful web query method that reduces research time, the WRDS Cloud for executing research and strategy development, and the WRDS client server using PCSAS, Matlab, Python and R.
Our Analytics team, doctoral-level support and rigorous data review and validation give clients the confidence to tailor research within complex databases and create a wide range of reliable data models.
WRDS provides access to S&P Capital IQ, CRSP, NYSE, Thomson Reuters, Bureau van Dijk, Global Insight, OptionMetrics and other important business research databases.
From partnerships with data vendors, to our own tools including the WRDS SEC Analytics Suite, WRDS Quant Alpha Platform and the Wharton School's OTIS - WRDS is the global gold standard in data management and research, all backed by the credibility and leadership of the Wharton School.

# WRDS Home Page



# Running the data query

WRDS provides many different databases with different types of data.

- Stock price data for US firms is provided by CRSP.
- To query the CRSP data, select CRSP under Your Subscriptions.
- Inside CRSP, click on Stock / Security Files
- Under Stock / Security Files, click on Monthly Stock File

# Step 1: Enter the Date Range

Query Form | Variable Descriptions | Manuals and Overviews | FAQs | Dataset List

## CRSP Monthly Stock

Click here to preview this dataset! (Beta)

**Step 1:** Choose your date range.

Date range

2001-01 to 2001-12

# Step 2: Select companies

**Step 2:** Apply your company codes.

◉ TICKER ○ PERMNO ○ PERMCO ○ CUSIP ○ NCUSIP ○ HSICCD ○ SICCD

Select an option for entering company codes

◉ IBM AMZN WMT

*Please enter Company codes separated by a space.*

*Example: IBM MSFT AAPL [ Code Lookup ]*

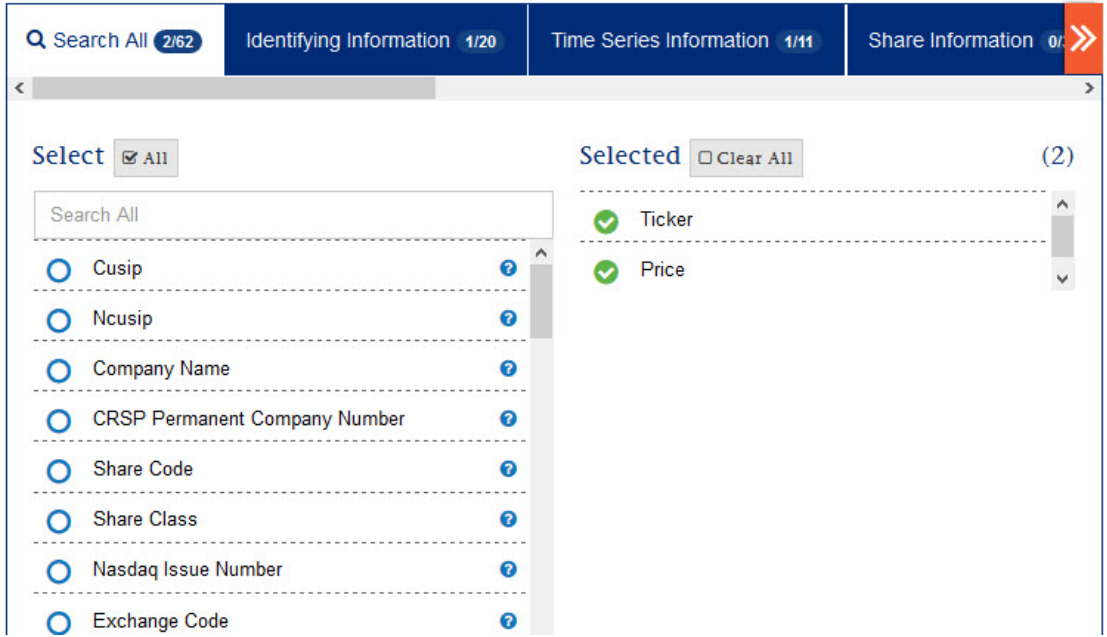
☐ Code List Name

*Save code list to Saved Codes*

# Step 3: Select the Variables

**Step 3:** Query Variables.

*How does this work?*



# Step 4: Select the Output Format

**Step 4:** Select query output.

Select the desired format of the output file. For large data requests, select a compression type to expedite downloads. If you enter your email address, you will receive an email that contains a URL to the output file when the data request is finished processing.

**Output Format**

- (●) fixed-width text (*.txt)
- ( ) comma-delimited text (*.csv)
- ( ) Excel spreadsheet (*.xlsx)
- ( ) tab-delimited text (*.txt)
- ( ) HTML table (*.htm)
- ( ) SAS Windows_32 dataset (*.sas7bdat)
- ( ) SAS Windows_64 dataset (*.sas7bdat)
- ( ) SAS Solaris_64 dataset (*.sas7bdat)
- ( ) dBase file (*.dbf)
- ( ) STATA file (*.dta)
- ( ) SPSS file (*.sav)

**Compression Type**

- (●) None
- ( ) zip (*.zip)
- ( ) gzip (*.gz)

# Step 5: Download after processing

Wharton University of Pennsylvania | wrds WHARTON RESEARCH DATA SERVICES | Search WRD | Get Data | Analytics | Cla

Home / Your Account / Queries / Query 8904795, for crsp_a_stock

## Query 8904795, for crsp_a_stock

| | |
|---|---|
| Status: | Success — Rerun |
| Product: | crsp_a_stock |
| Results: | Result Size: 15 KiB<br>Result Count: 240 Rows<br>Output Files:<br>Download .xlsx Output<br>Result data were retained for 2 days and are no longer available. |
| Timing: | Elapsed Time: 4 seconds<br>Submitted: 2024-09-19 07:19<br>Work Begun: 2024-09-19 07:19<br>Work Finished: 2024-09-19 07:19 |
| Input Parameters: | Toggle Input Parameters |

# OBTAINING STOCK RETURNS

# Stock returns

- Return = sum of cash flows received over a given horizon, relative to today's purchase price

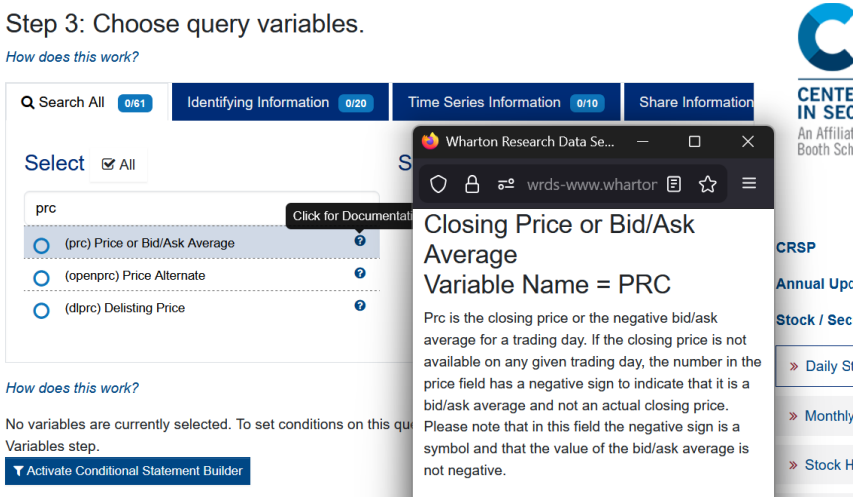
$$R_t = \frac{P_t - P_{t-1} + D_1}{P_{t-1}}$$

- Q: What happens to prices around dividend dates?

- Beware of stock splits! With a x for y stock split, compute returns as: $R_t = \frac{\frac{x}{y}P_t - P_{t-1} + \frac{x}{y}D_1}{P_{t-1}}$
  - E.g. "5 for 4 stock split": x/y = 5/4 = 1.25

# Stock returns via CRSP

- CRSP variable “prc” gives closing price
- CRSP variable “divamt” gives dividend
- CRSP variable “ret” gives return
  - Over given holding period (check frequency: daily, monthly, quarterly, annually)
  - Conveniently adjusts for stock splits, irregular dividend payments, potential delisting returns ect!



# CLASS ROOM EXERCISE: INDUSTRY CHARACTERISTICS

# How are “industries” defined?

- It’s complicated! (it’s life)
  - What if a firm changes over time?
  - What if a firm does multiple things?
  - What if an industry did not even exist last year?
  - What about tech firms? Is Amazon a book store? A retailer? A computer processing service firm? “Information Technology”? “Consumer Discretionary”?
- There are many! Important ones you will encounter include:
  - SIC – *Standard Industrial Classification. E.g. SIC code 2024 “ice cream and frozen desserts”*
  - NAICS – *North American Industry Classification System. E.g. NAICS code 6212 “offices of dentists”*
  - FAMA FRENCH 5/10/12/17/39/38/48/49. *E.g. Fama French 37 “Chips” Electronic Equipment*
  - GICS *Global Industry Classification Standard*. E.g. *GICS 25102020 “Motorcycle Manufacturers”*
  - NACE *Statistical Classification of Economic Activities in the European Community*. E.g. *NACE 07.21 “Mining of uranium and thorium ores”*

# Example: Fama/French 49 industries

- Universe of firms with publicly listed stock, as of Dec2018 (CRSP)
- Details and background
  - "FUN" is Entertainment
  - "LABEQ" is Measuring and Control Equipment
  - …
- Why are there such large differences in number of firms across industries? Bad categories?

| Date | Fama and French 49 Industries - Description | NFIRM | | Fama and French 49 Industries - Description | NFIRM |
|---|---|---|---|---|---|
| 31DEC2018 | AERO | 16 | | INSUR | 96 |
| 31DEC2018 | AGRIC | 6 | | LABEQ | 53 |
| 31DEC2018 | AUTOS | 51 | | MACH | 96 |
| 31DEC2018 | BANKS | 437 | | MEALS | 59 |
| 31DEC2018 | BEER | 11 | | MEDEQ | 124 |
| 31DEC2018 | BLDMT | 55 | | MINES | 16 |
| 31DEC2018 | BOOKS | 17 | | OIL | 136 |
| 31DEC2018 | BOXES | 10 | | OTHER | 21 |
| 31DEC2018 | BUSSV | 163 | | PAPER | 26 |
| 31DEC2018 | CHEMS | 66 | | PERSV | 30 |
| 31DEC2018 | CHIPS | 153 | | RLEST | 30 |
| 31DEC2018 | CLTHS | 30 | | RTAIL | 140 |
| 31DEC2018 | CNSTR | 44 | | RUBBR | 17 |
| 31DEC2018 | COAL | 8 | | SHIPS | 9 |
| 31DEC2018 | DRUGS | 486 | | SMOKE | 3 |
| 31DEC2018 | ELCEQ | 48 | | SODA | 10 |
| 31DEC2018 | FABPR | 5 | | SOFTW | 281 |
| 31DEC2018 | FIN | 104 | | STEEL | 28 |
| 31DEC2018 | FOOD | 41 | | TELCM | 72 |
| 31DEC2018 | FUN | 46 | | TOYS | 21 |
| 31DEC2018 | GOLD | 9 | | TRANS | 68 |
| 31DEC2018 | GUNS | 7 | | TXTLS | 7 |
| 31DEC2018 | HARDW | 39 | | UTIL | 76 |
| 31DEC2018 | HLTH | 51 | | WHLSL | 96 |
| 31DEC2018 | HSHLD | 38 | | | |
| | | | | TOTAL | 3456 |

# Characterizing industries

| | | |
|---|---|---|
| Asset Turnover | Efficiency | Sales as a fraction of the average Total Assets based on the most recent two periods |
| Inventory Turnover | Efficiency | COGS as a fraction of the average Inventories based on the most recent two periods |
| Payables Turnover | Efficiency | COGS and change in Inventories as a fraction of the average of Accounts Payable based on the most recent two periods |

Considering the Fama-French 49 industries, which industries would you expect to have…

- High inventory turnover?
- Low inventory turnover?

Let's take a look: *Financial Ratios Suite by WRDS*

# Your assignment

Choose

- Financial Ratios Industry Level
- Step 1 Date: 2018-01-01 to 2018-12-31
- Step 2 Industry classification: Fama French 49
- Step 3 Variables: Inventory Turnoverv “inv_turn”
- Step 4 Aggregation: Median
- Step 6 Output: comma-delimited text, csv

Q: Industries with highest Inventory Turnover?

Q: Industries with lowest Inventory Turnover?

# Preparation for upcoming case study

— What do these firms do? How does it show in their financials? (Check via WRDS)

1. Aircraft manufacturer
2. Airline
3. Bookstore chain
4. Commercial bank
5. Computer software developer
6. Department store chain
7. Electric and gas utility
8. IT hardware and storage provider
9. Online retailer
10. Parcel delivery service
11. Pharmaceutical company
12. Retail grocery chain
13. Software as a service provider for customer relationship management
14. Social networking service